

Exam Signal Processing

Winter Term 20.. - March ..nd, 20..

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Name _____ Student Number _____

Signature _____

- The exam comprises **10 problems** and a total of **17 pages**. Credit for each question is proportional to the amount of time you should spend on it. You have **120 minutes** to complete this examination.
- You are allowed to use a non-programmable calculator.
- Write your **name** and **student number** on **each** page, including additional sheets of paper if provided by the examiner. Make sure that additional sheets of paper are numbered accordingly, too.
- Make sure to use an **indelible** pen with either **blue** or **black** ink. We will **not** grade any question worked in red pen, fountain pen or pencil.
- Only students on the official list provided by the Examination Office are admitted to the examination. The Examination Office reserves the right to deprive any admittance at a later time for all students not listed.
- Attempted deception, use of unauthorized materials or prohibited resources will result in a grading of the exam as **failed (5.0)**. This includes but is not limited to any kind of communication with another student during the examination and the possession of unauthorized materials before distribution and the duration of the exam. Please note that you will also need to turn off your phone for the entire duration of the exam. Grades will remain subject to be disallowed in cases attempted deception is recognized at a later time.
- **By taking the exam**, you acknowledge to not suffer from any health impairments that might reduce your performance ability during the test. In case you fall ill during the exam, you can withdraw from the exam and terminate it. In such a case, you will need to report your illness to the supervisor and leave the room. You will need to provide a medical certificate, which includes the exact date, time and the symptoms of your illness and to what extent this temporarily prevents you from writing an examination. An insufficient description of your inability to take the exam or an incapacity certificate will not be accepted. The medical examination also needs to take place on the very same day of your withdrawal. Moreover, the medical certificate needs to be submitted immediately after the medical examination to the Examination Office.
- By signing the form, you agree on all terms mentioned before. Please also note that this signature is required to be admitted to the exam.

Q1.1 (X)	Q1.2 (X)	Q1.3 (X)	Q1.4 (X)	Q2 (X)	Q3.1 (X)	Q3.2 (X)	Q4.1 (X)	Q4.2 (X)	Q4.3 (X)	Σ (X)	Grade

Date, Signature of Examiner

Exam was inspected (to be filled out by student):

Date _____ Student's Signature _____

Question 1**(X points)**

Determine if the provided systems $\mathfrak{S}(x(t)):x(t) \rightarrow y(t)$ are linear, causal or time-invariant. Please make sure that you explain/prove each of your conclusion.

a) $y(t) = x^2(t)$.

b) $y(t) = x(t) \cdot \sin(\omega t)$.

c) $y(t) = \dot{x}(t)$.

Question 2**(X points)**

Calculate the cross-correlation of $x_1[n] = [2, 1, 2, 1]$ and $x_2[n] = [1, 2, 3, 4]$.

Question 3**(X points)**

We would like to acquire a sinusoidal signal with its frequency $f_s = 20$ Hz. Determine

- the sampling frequency f_A
- and the corresponding sampling interval T_A

to avoid aliasing.

Please state each of your conclusions by using a relational operator (e.g., $<$, $>$, $\leq$, $\geq$, $=$).

Question 4**(X points)**

Express **algebraically** the relationship between the impulse response $h(t)$ and step response $y_{\sigma}(t)$.

Question 5**(X points)**

One way to design IIR filters is to strategically place poles and zeros in the z -plane.

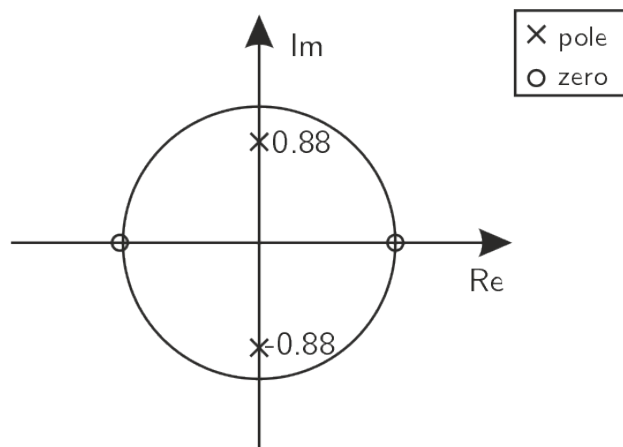


Figure 1: Pole-Zero-Diagramm.

a) Determine the transfer function $H(z)$ of the digital filter with the help of Figure 1.

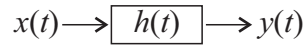
b) Evaluate the transfer function $H(z)$ for $z = e^{jk\frac{\pi}{2}}$ at $k[n] = [0, 1, 2, 3, 4]$.

c) Sketch the frequency response $|H(e^{j\omega})|$.

d) Determine the corresponding difference equation and sketch a block diagram, which represents the filter.

Question 6**(X points)**

A causal, time-continuous system is given by following block graph:

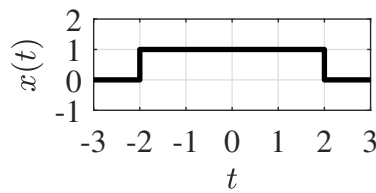


Its impulse response $h(t)$ is defined as

$$h(t) = \alpha \cdot (\sigma(t) - \sigma(t - 2)),$$

with $\alpha \in \mathbb{R}^+$.

We will consider that the depicted signal $x(t)$ is applied to the input of the system.



a) Draw the signal $h(t)$ in the range of $-4 \leq t \leq 4$. Make sure that axes are labeled correctly.

b) The convolution of the two signals $x(t)$ and $h(t)$ can be split up to three non-zero subintegrals:

Subintegral I, valid in time interval $t_1 \leq t < t_2$: $y_I(t) = \int_a^b \alpha \, d\tau$

Subintegral II, valid in time interval $t_2 \leq t < t_3$: $y_{II}(t) = \int_c^d \alpha \, d\tau$

Subintegral III, valid in time interval $t_3 \leq t < t_4$: $y_{III}(t) = \int_e^f \alpha \, d\tau$

Specify their parameters $t_1, t_2, t_3, t_4, a, b, c, d, e, f$.

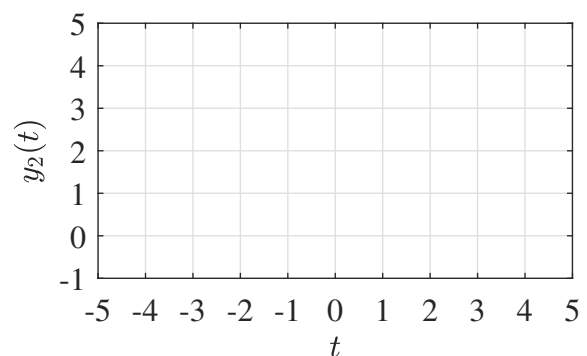
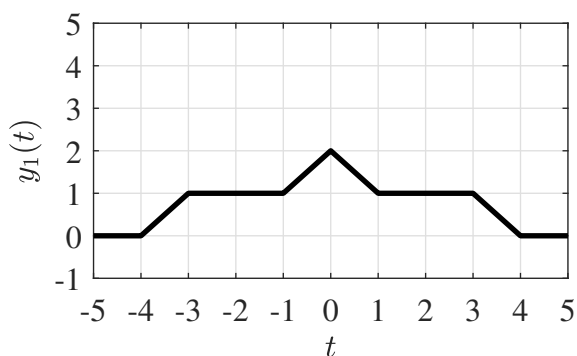
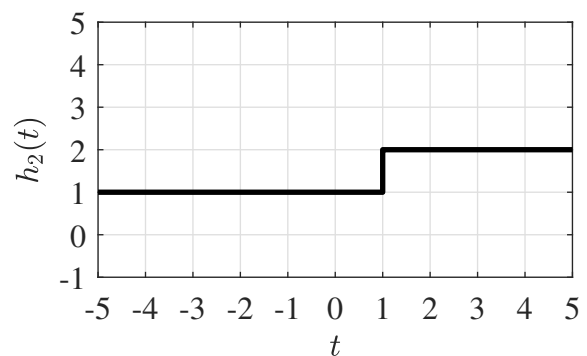
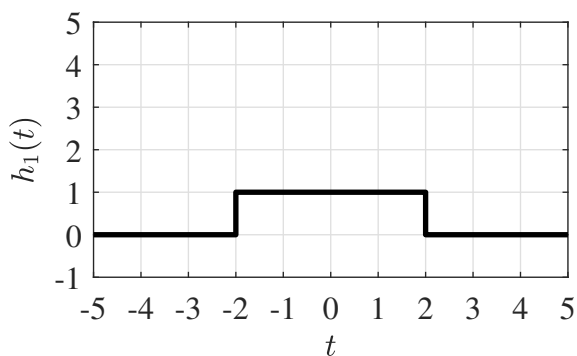
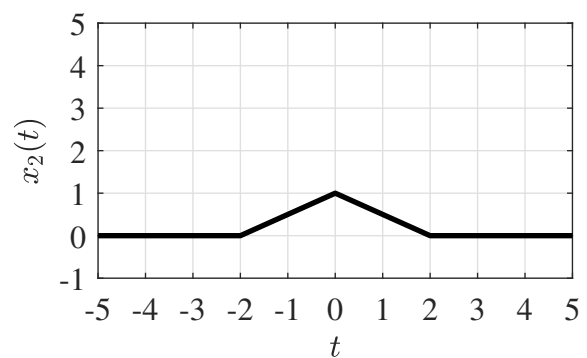
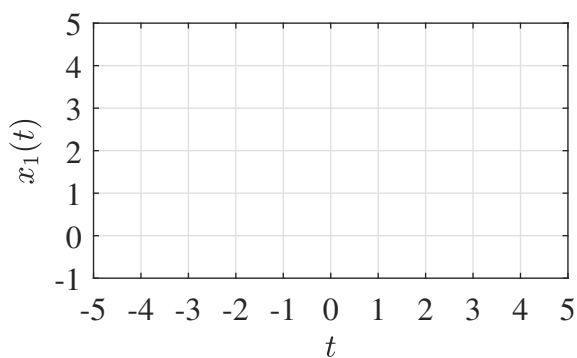
Question 7

(X points)

The convolution

$$y(t) = x(t) * h(t)$$

of two input signals $x_{1-2}(t)$ and two impulse responses $h_{1-2}(t)$ was calculated, respectively. The results of each convolution $y_{1-2}(t)$ is depicted below. Sketch the respective missing signal in the range of $-5 \leq t \leq 5$.



Question 8**(X points)**

The magnitude $|Y(j\omega)|$ and phase $\phi(j\omega)$ of several Fourier Transforms (1-6) are shown on the following page. Their corresponding time functions are depicted (A-F) as well.

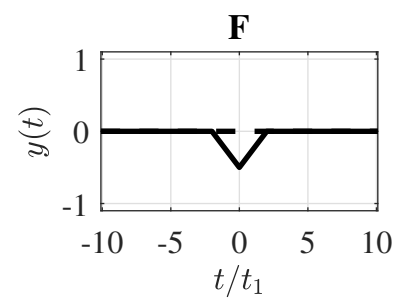
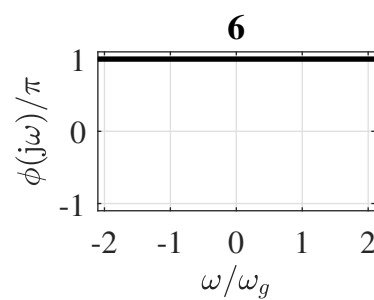
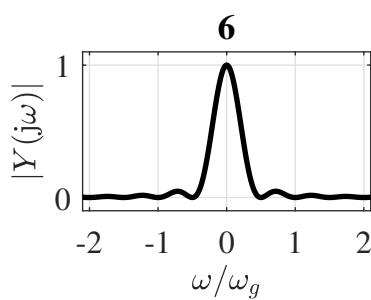
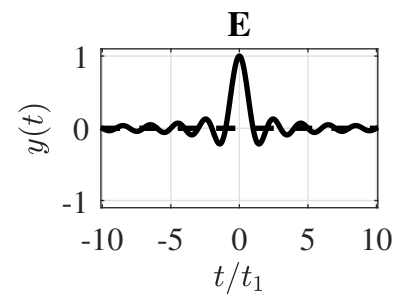
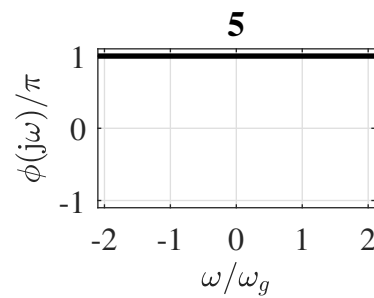
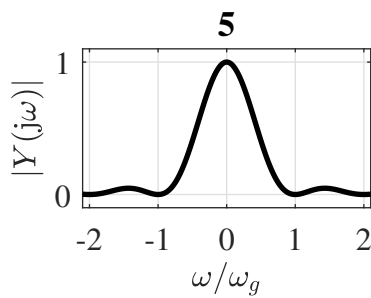
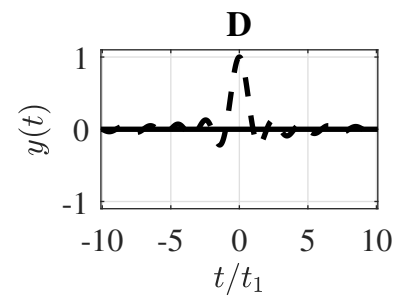
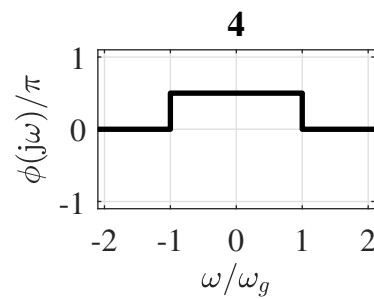
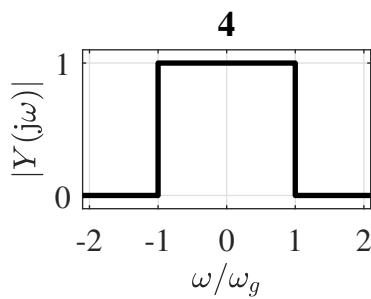
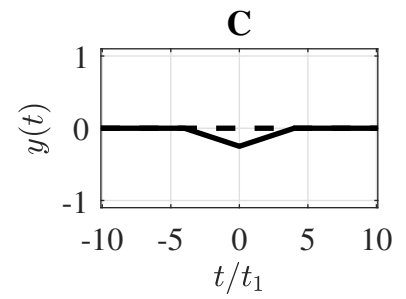
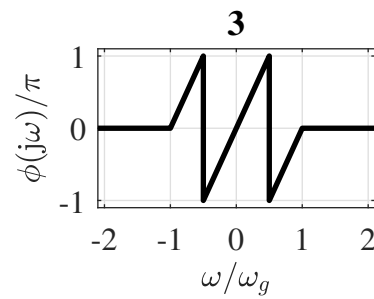
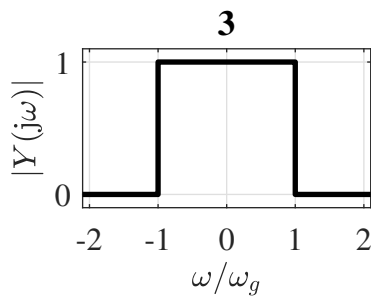
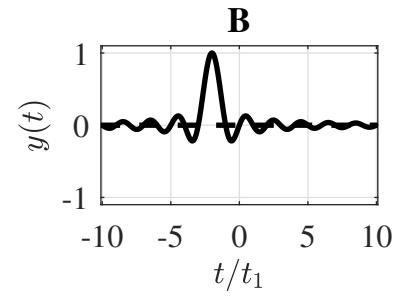
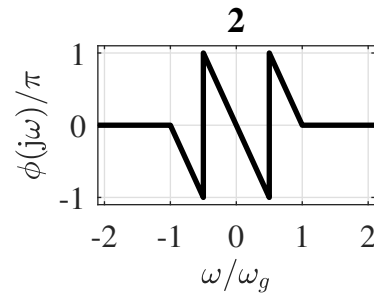
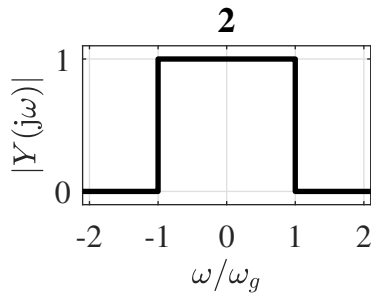
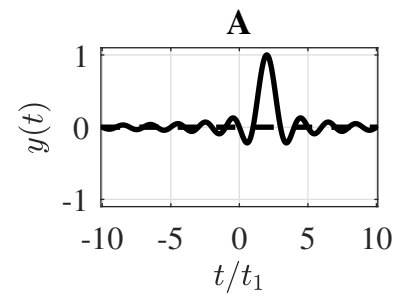
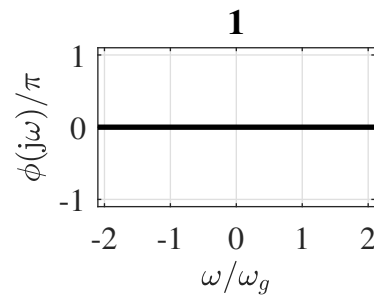
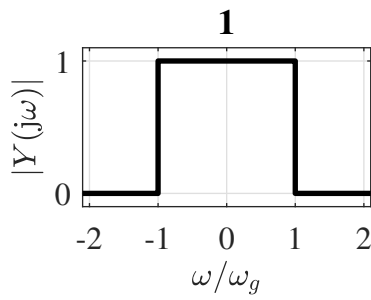
$$Y(j\omega) = |Y(j\omega)| \cdot e^{j\phi(j\omega)} \bullet \text{---} \circ y(t).$$

These time functions are characterized by their real part (—) and imaginary part (- - -), for which the relation

$$t_1 = \frac{\pi}{\omega_g}$$

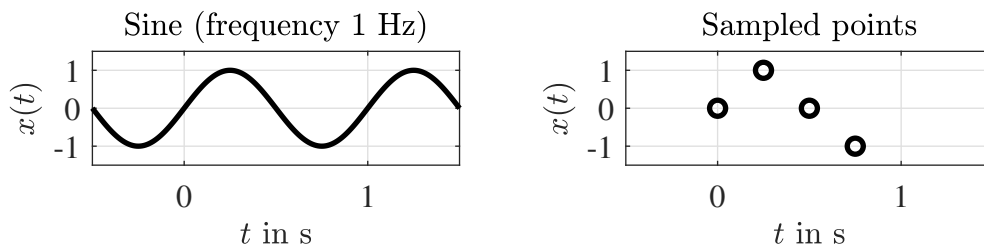
is exploited. Unfortunately, the corresponding time functions (A-F) are not in the right order anymore. Assign each Fourier Transform (1-6) to their corresponding time function (A-F) and briefly explain your choice.

For instance: H-7, because ...



Question 9**(X points)**

We acquire a sine signal with its frequency $f_{\text{sig}} = 1 \text{ Hz}$ by using a sampling frequency $f_{\text{Samp}} = 4 \text{ Hz}$.



As a result, we receive 4 sample points. Thus, our measured discrete-time signal comprises the values

$$x[n] = [0, 1, 0, -1].$$

- a) Determine the 4-point discrete Fourier transform (DFT_4) $X[m]$ of the discrete-time signal $x[n]$.

- b) State the formula, which is used for calculating the frequency resolution of the discrete Fourier Transform. Then, determine the frequency resolution for DFT_4 . Make sure you also assign the corresponding frequency to each value of $X[m]$.
- c) Sketch the magnitude and phase of $X[m]$ as a function of m .
- d) Briefly describe (without any calculation) how the magnitude and phase of $X[m]$ would change if you acquired a cosine ($x[n] = [1, 0, -1, 0]$) instead.

Question 10**(X points)**

We would like to design an IIR low-pass filter, which will be used in a mobile phone to separate the human voice from ambient noise. This lowpass filter shall meet the following specifications:

- Maximum passband attenuation: 3 dB
 - Minimum stopband attenuation: 20 dB
 - Passband edge: 4 kHz
 - Stopband edge: 8 kHz
 - Sampling frequency: 20 kHz
- a) Draw the tolerance bounds on the magnitude of the frequency response $|H_C(j\omega)|$. For this purpose, calculate the corresponding digital filter specifications.

- b) Determine the minimum order N and cutoff frequency Ω_C such that the butterworth filter meets the mentioned specifications. Make sure that the stopband edge is met exactly.

- c) Determine $H(z)$ of the discrete time filter by applying the bilinear transform to the continuous time prototype filter

$$|H_C(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_C)^{2N}}.$$

Use $T_d = 2$.

- d) What are the main advantages of using an IIR filter over an FIR filter?
- e) If the transition width were increased in the design specification, would the required filter order N increase or decrease?
- f) If the specification for the minimum stopband attenuation is 40 dB instead of 20 dB, will the required filter order N increase or decrease?